Less to Lose:

How Shorter and More Flexible Degrees Shape University Choice and Earnings

Emma Scandolo

University College London

November 5, 2026

How long should you study until your first degree?

- Length and flexibility of degrees vary
  - **Between** countries: economics 3 yrs England, 4 US, 4–5 LatAm
  - **Within** countries: economics 3+2, law 5

- What are the costs and benefits of each design?
  - Shorter first degree: lower cost, but less training
  - Intermediate credential: less to lose, but early stopping

- **Bologna Reform** (1999): 20+ countries adopt common BA + MA structure
  - A large natural experiment to study these trade-offs

A heated (old) debate

Expresso

02/00/0 actualizado às 0:00

Bologna is a "gigantic fraud"





Yet still relevant nowadays: Latin America deciding now

Argentina

LOS ANDES

The Andes › society › economy December 13, 2025 - 08:10

UNCuyo's Economic Sciences careers will be shorter: how it will affect students

Public Accountant, Administration, Economics and Logistics will have new curricula that reduce their duration to four years and incorporate intermediate degrees after two years to favor early labor insertion.

Chile

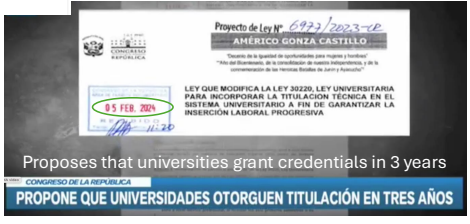
emol. NATIONAL emol.social

With definition of programs and legal adjustments: Avanza plan that would shorten university careers

Both the Commission of Academic Vice-Rectors of the CRUCh and the Aequalis forum discuss the idea, which has gained strength in recent months.

June 29, 2026 | 09:23 | Emol33

Peru



This paper

- **Design**: RD across birth cohorts in Portugal; Italy as a check
- **Reduced-form effects**, Portugal
  1. **+0.23** (0.06)
     ↑ years of university

     +2.1% (0.5)
     ↑ hourly wages
  2. **+2.3 pp** (0.4)
     ↑ degrees in fields with an intermediate credential (IC)

     +1.06 pp (0.31)
     ↑ full five-year completion in fields with IC
- **Model**: students learn by trying. An intermediate credential inside a long degree means less to lose, more students try and some finish the full degree

Contribution

- **Design of higher-education systems** (Bordon and Fu, 2015; Hastings et al., 2013; Kirkeboen et al., 2016; Mountjoy, 2022; Angrist et al., 2022)
  → A new aspect: length and flexibility

- **Community colleges and option value models** (Rouse, 1995; Mountjoy, 2022; Manski, 1989; Stange, 2012; Trachter, 2015; Arcidiacono et al., 2025)
  → Model estimated to match the moments of a reform that **adds** option value

- **Bologna reform literature** (Cappellari and Lucifora, 2009; Kroher et al., 2021)
  → First comprehensive paper to document its effects

Background

Two reforms within one

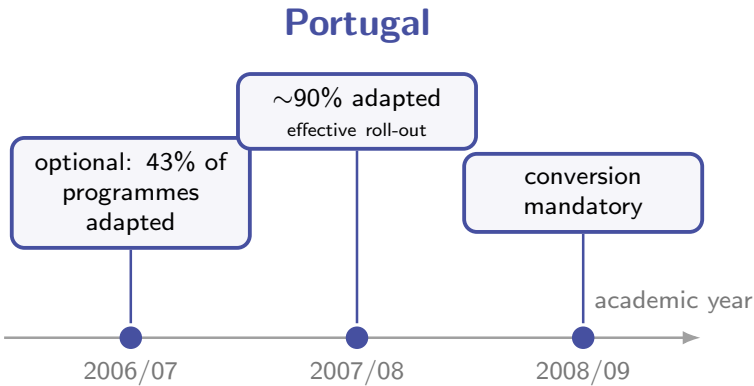
Portugal

	pre	post	
Engineering, computing & architecture	5	**3+2**	new intermediate credential
Economics, sciences, humanities, social sciences	4(+2)	**3+2**	shorter first-cycle degree
Nursing, medicine, law	4–6	4–6	no change

Italy: only $5 \to 3{+}2$ or $4 \to 3{+}2$

What the 3-year degree is worth

Implementation timing



- **Italy**: optional adoption in 2000/01, mandatory in 2001/02

Italy: early vs late regions | Capacity and age at entry

Data and RD design

Data

1. **Quadros de Pessoal** (QP): 2000–2023
   - ▷ Linked employer–employee data covering the private sector
   - ! *Conditional* on private-sector employment

2. **Labour Force Survey** (LFS): 1998–2024
   - ▷ Repeated cross-section
   - ! *Unconditional* on employment type

3. **Education and Training Survey** (IEFA): 2007–2022
   - ▷ Dropout
   - ! *Small sample size*

Data

1. **Quadros de Pessoal** (QP): 2000–2023
   - ▷ Linked employer–employee data covering the private sector
   - ! *Conditional* on private-sector employment

2. **Labour Force Survey** (LFS): 1998–2024
   - ▷ Repeated cross-section
   - ! *Unconditional* on employment type

Italy external validity
LFS: 2009–2020

3. **Education and Training Survey** (IEFA): 2007–2022
   - ▷ Dropout
   - ! *Small sample size*

RD design

- Running variable: $Z_i = c_i - 2007$, with c_i the year i turns 18
- Treatment: $D_i = \mathbf{1}[Z_i \geq 0]$
- Reform effect τ = jump at the cutoff:

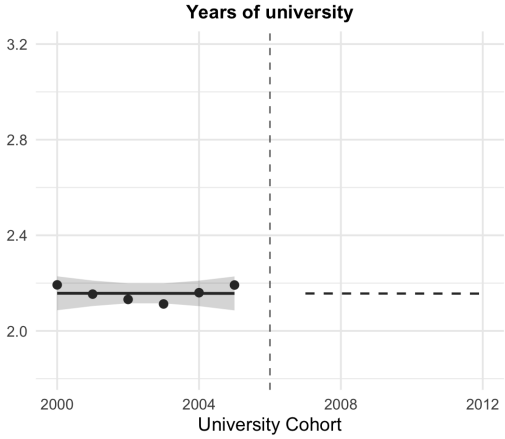
$$Y_i = \alpha + \tau D_i + \beta_1 Z_i + \beta_2 D_i Z_i + \varepsilon_i$$

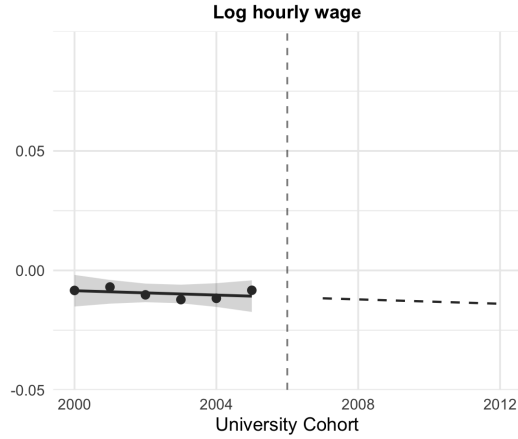
- Bandwidth: ± 6 cohorts; 2006 cohort excluded (donut)
- Education measured at 28, wages at 28–30

Why ages 28–30 | Wages: relative to ages 40–45 | Density | No choice delay

Does the reform increase attainment and pay?

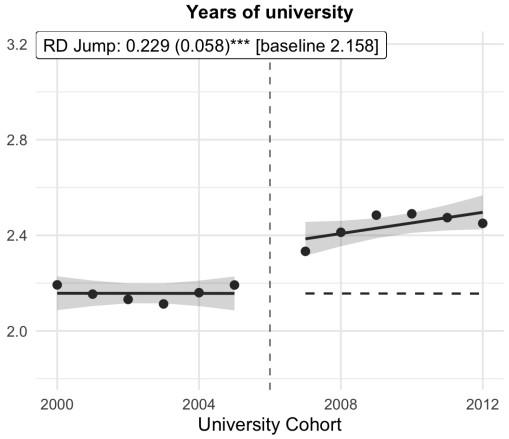
↑ years of university and ↑ hourly wages

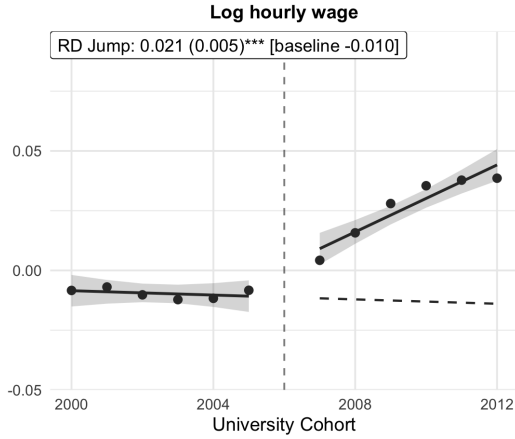




Women and men | Other ages and windows | Placebo cutoffs | Wages: relative to ages 40–45 | AKM

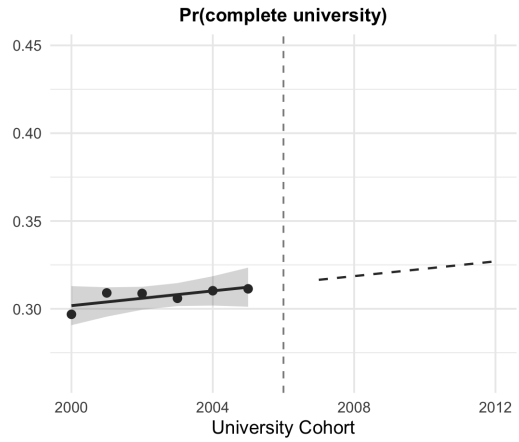
↑ years of university and ↑ hourly wages

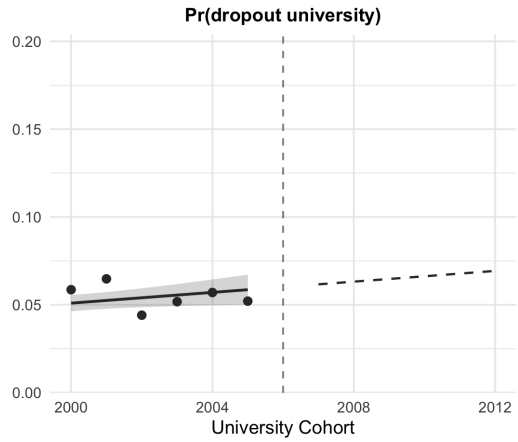




Women and men | Other ages and windows | Placebo cutoffs | Wages: relative to ages 40–45 | AKM

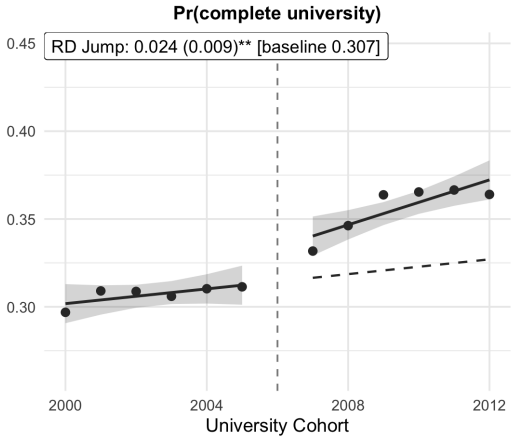
↑ university completion and ↑ dropout

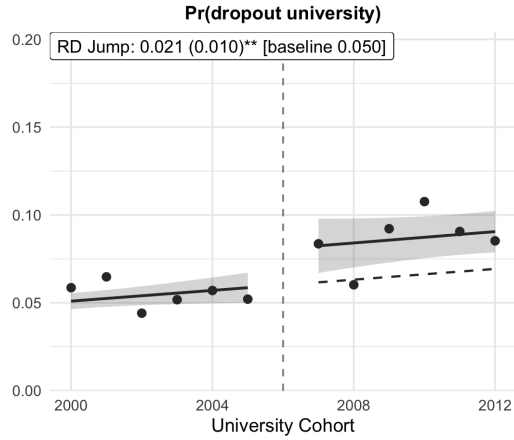




Women and men | QP vs LFS | Placebo cutoffs

↑ university completion and ↑ dropout





Women and men | QP vs LFS | Placebo cutoffs

Earnings: ↓ in the 20s, caught up by the early 30s

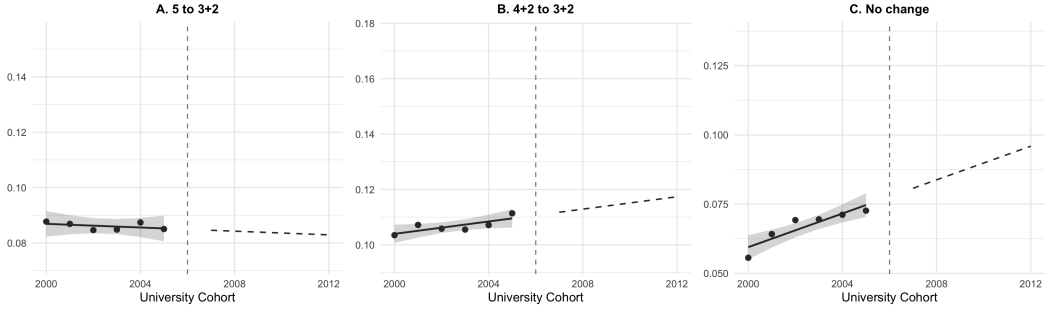


Women and men | AKM

Does the reform change the choice of field?

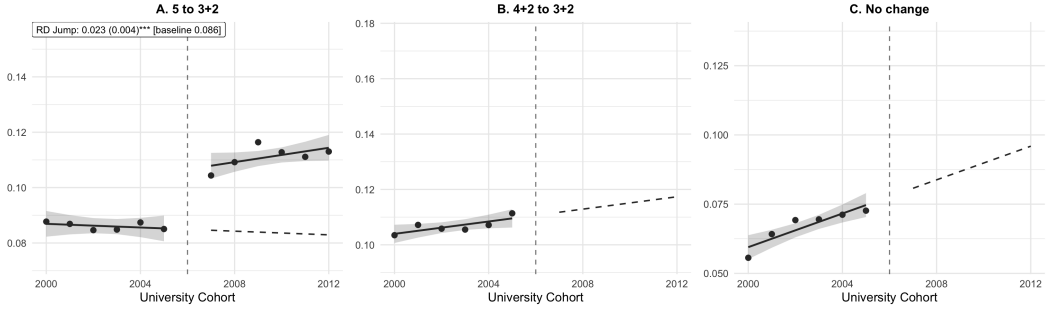
Do the changes carry into occupations and sectors?

↑ reformed fields, ↓ unchanged fields



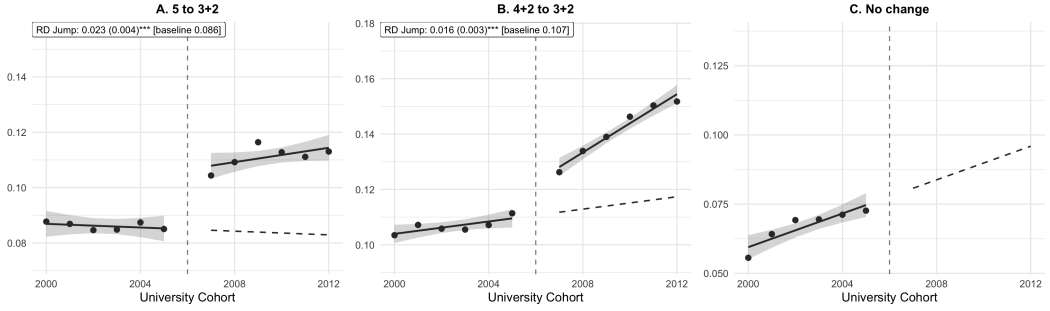
Women and men | Detailed fields | QP vs LFS | Applications | Placebo cutoffs

↑ reformed fields, ↓ unchanged fields



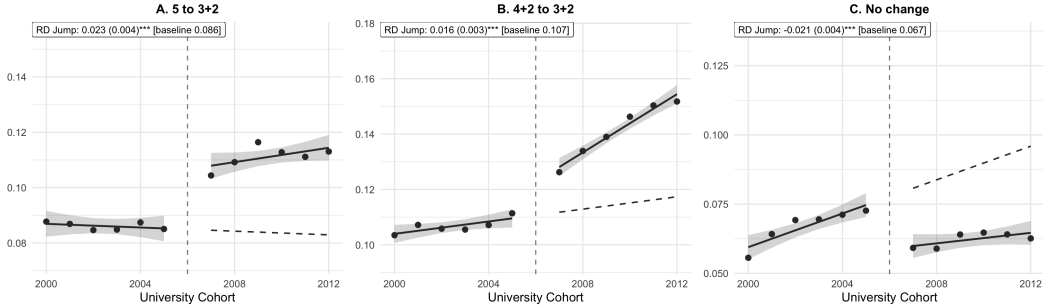
Women and men | Detailed fields | QP vs LFS | Applications | Placebo cutoffs

↑ reformed fields, ↓ unchanged fields



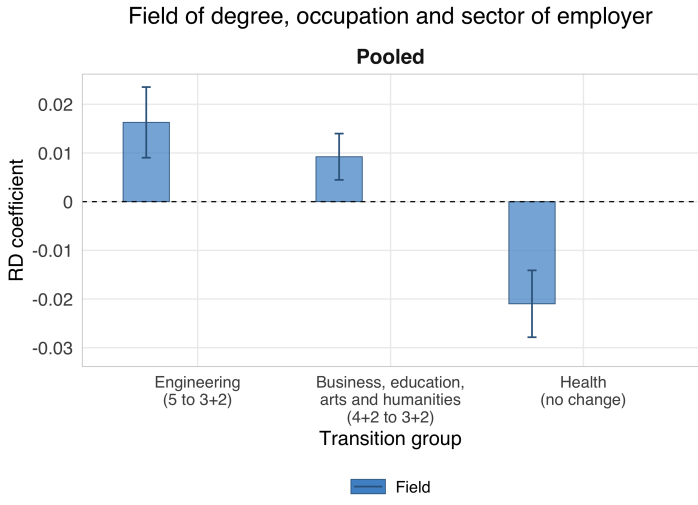
Women and men | Detailed fields | QP vs LFS | Applications | Placebo cutoffs

↑ reformed fields, ↓ unchanged fields



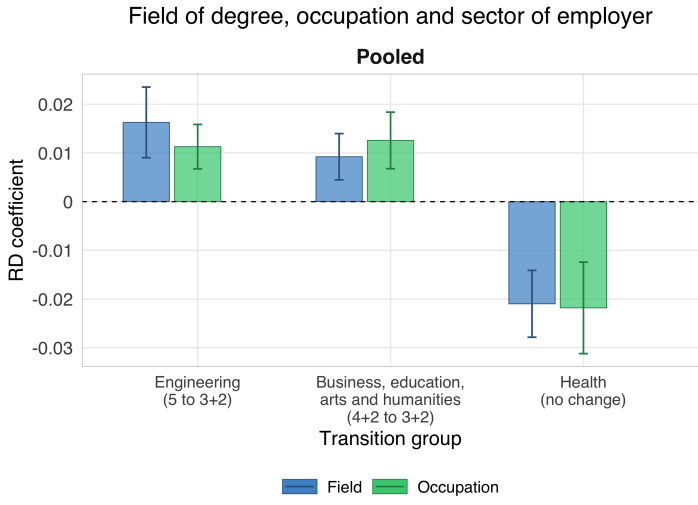
Women and men | Detailed fields | QP vs LFS | Applications | Placebo cutoffs

The shift carries into occupations and sectors



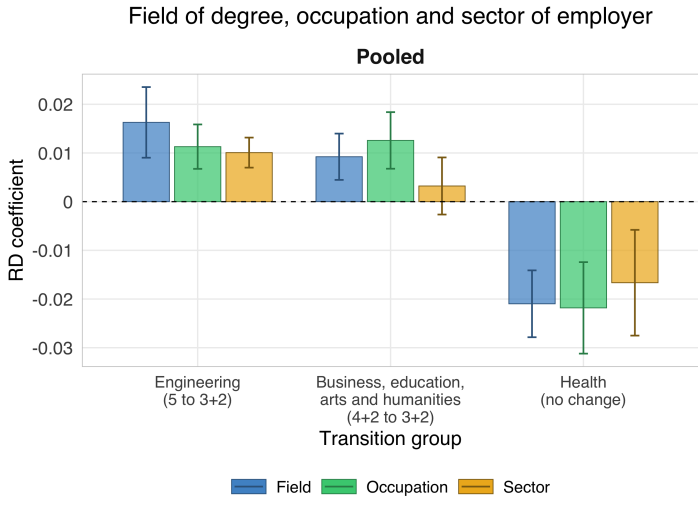
Women and men

The shift carries into occupations and sectors



Women and men

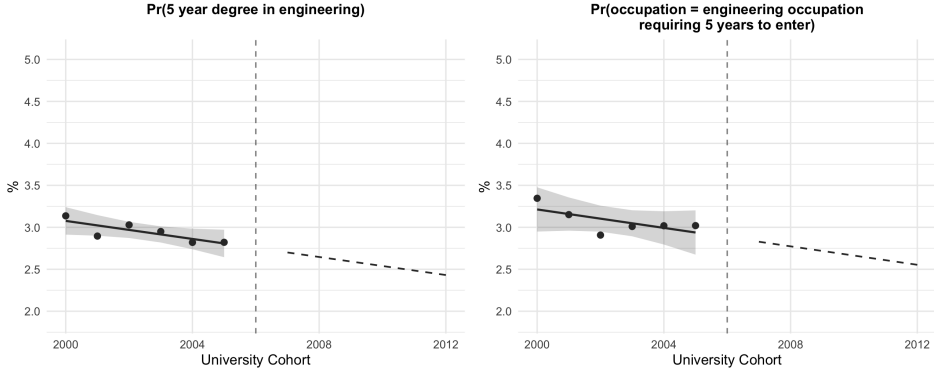
The shift carries into occupations and sectors



Women and men

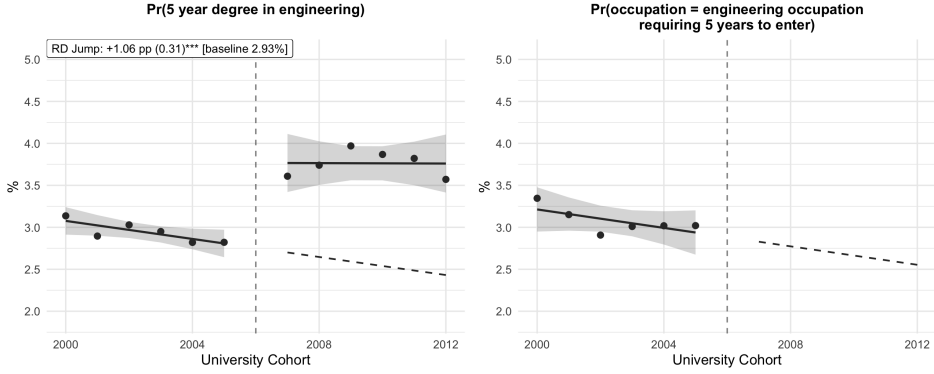
Does the 3-year credential divert students from completing the full degree?

Students are not just stopping at the new 3-year degrees



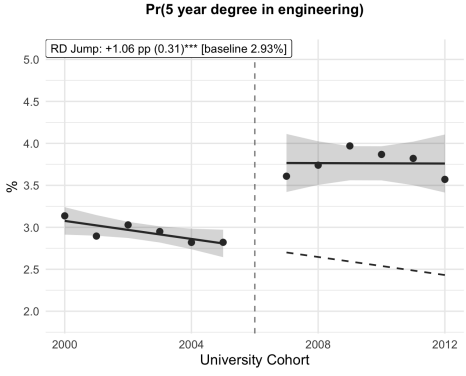
Women and men

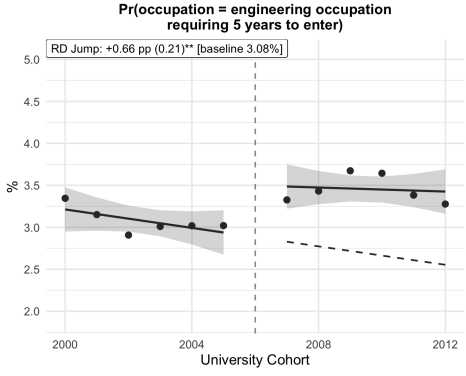
Students are not just stopping at the new 3-year degrees



Women and men

Students are not just stopping at the new 3-year degrees

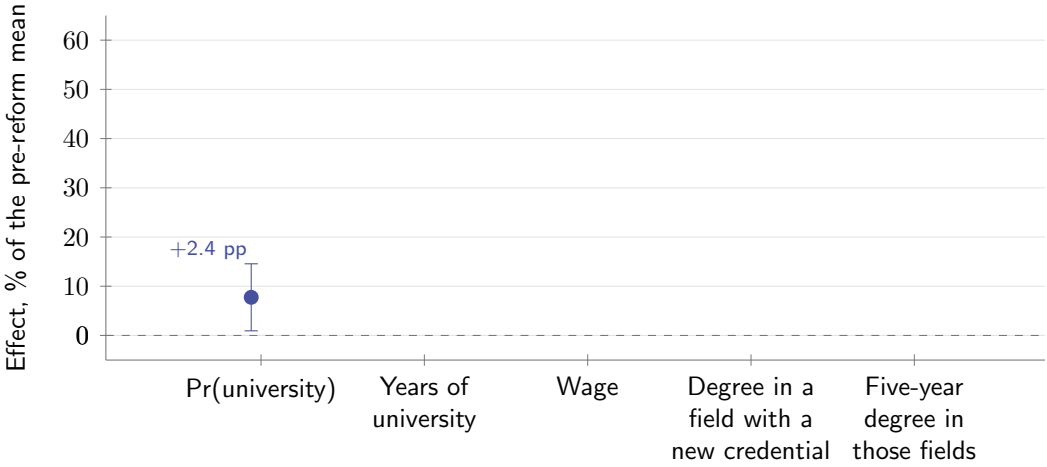




Women and men

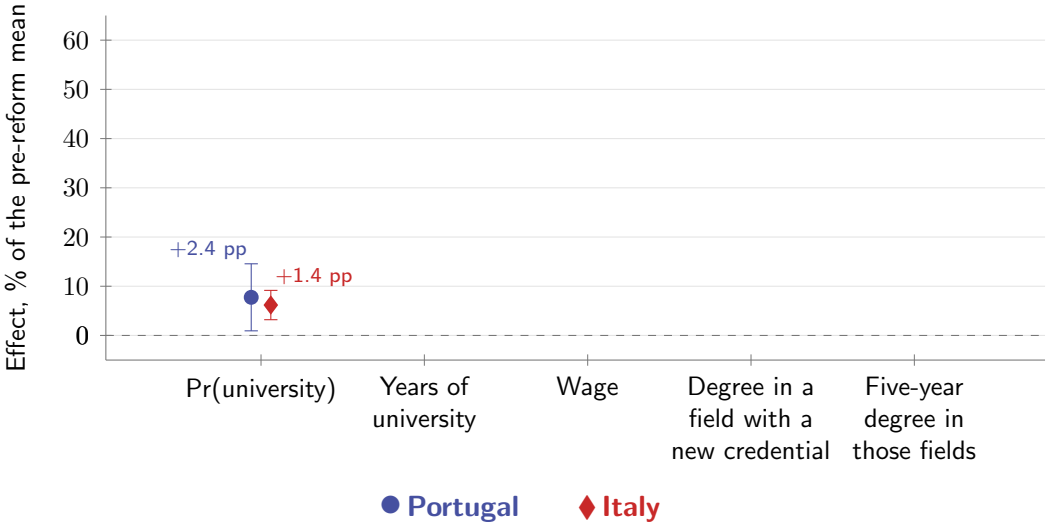
Does the same happen in Italy?

Portugal and Italy: the same pattern



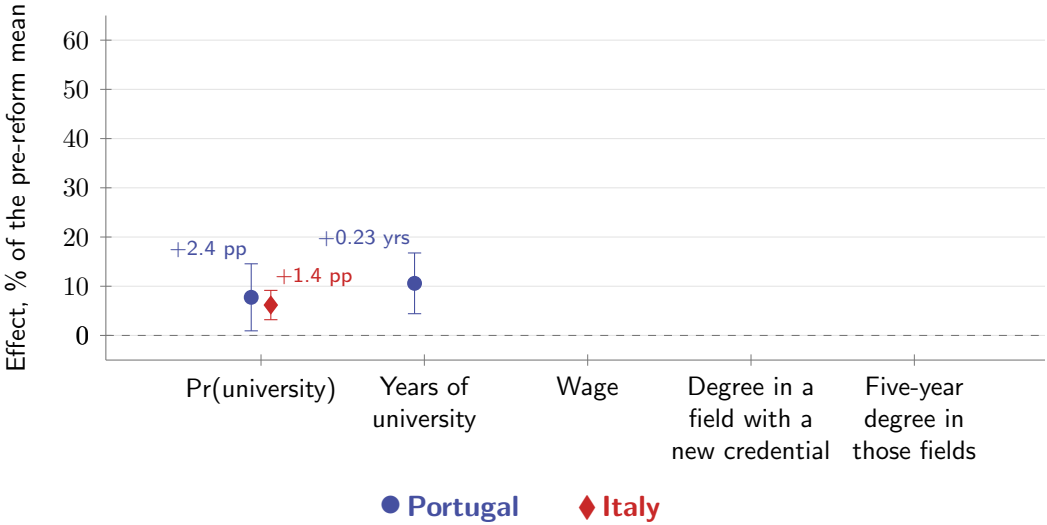
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



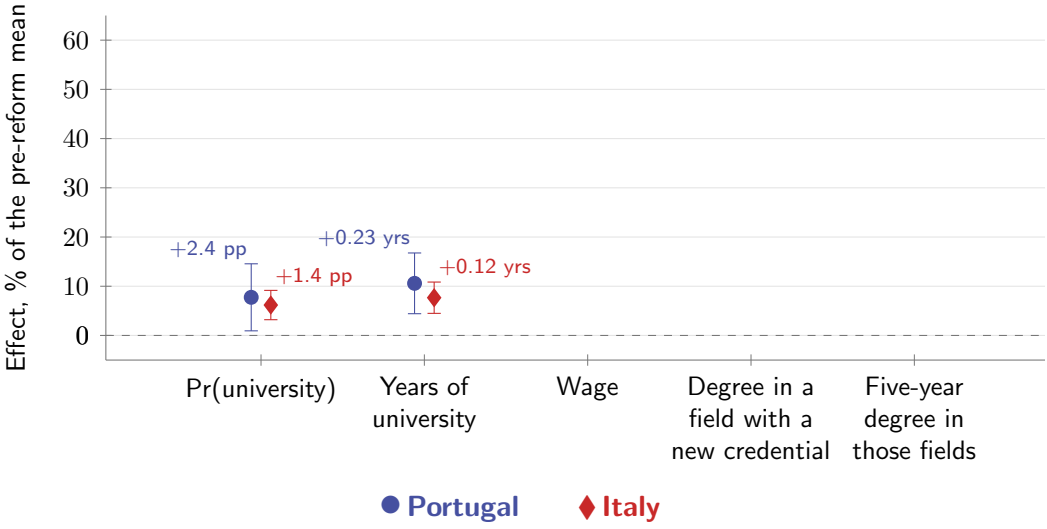
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



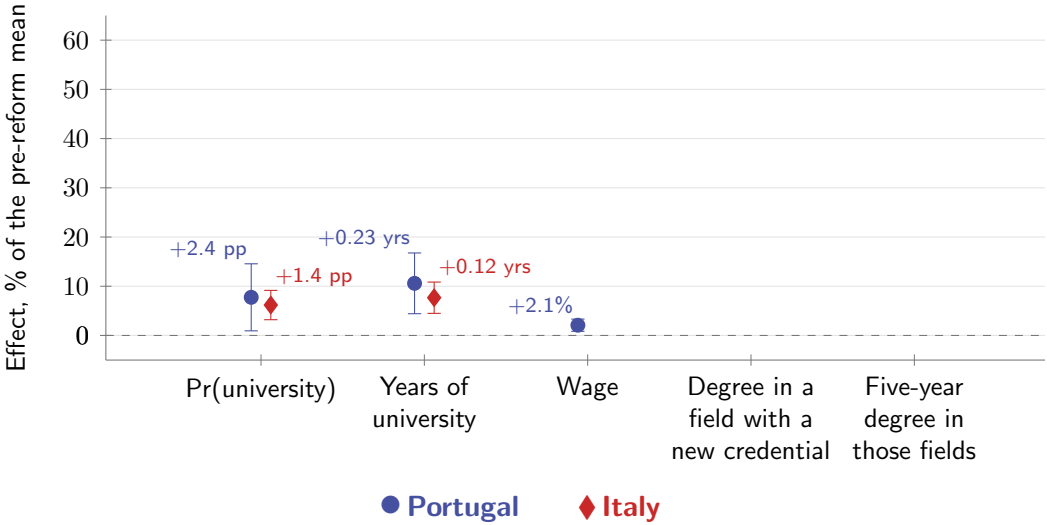
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



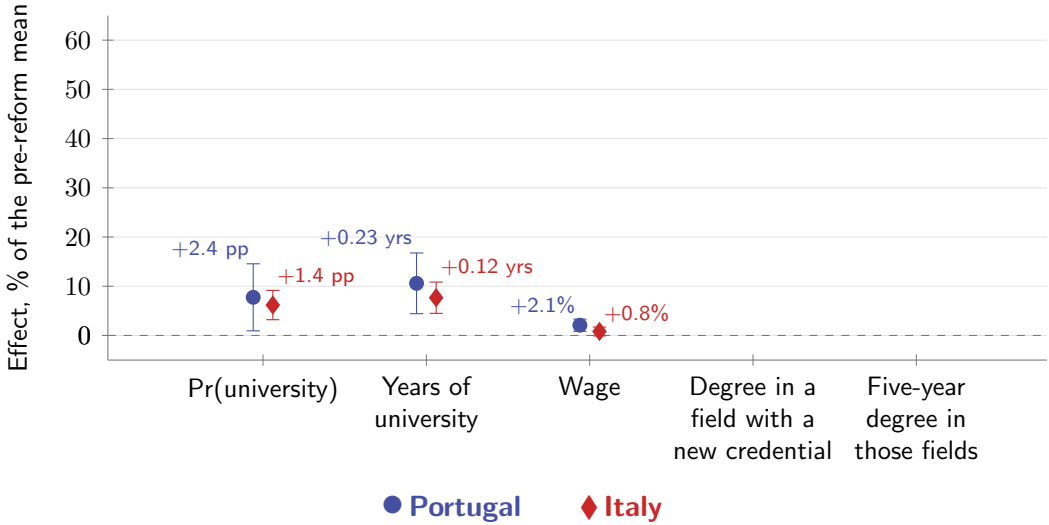
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



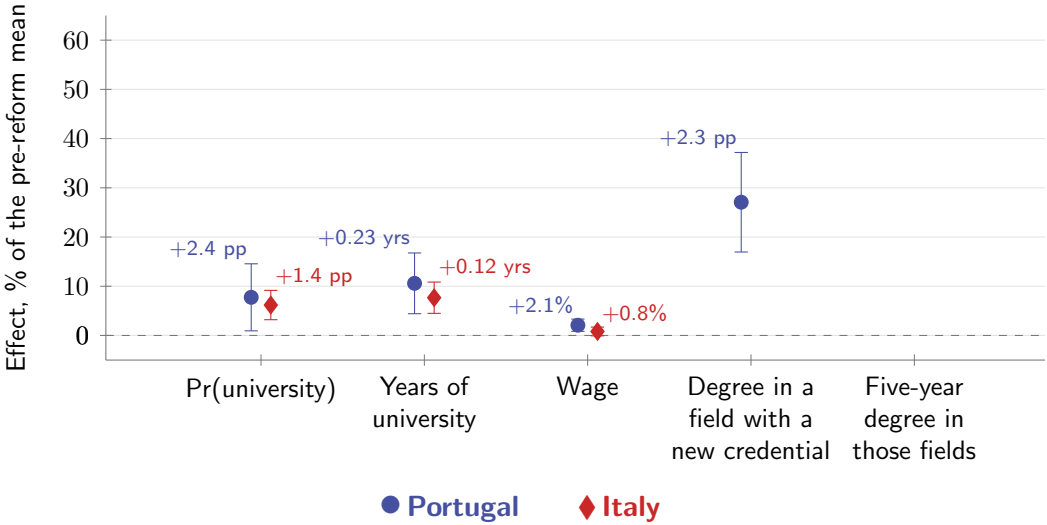
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



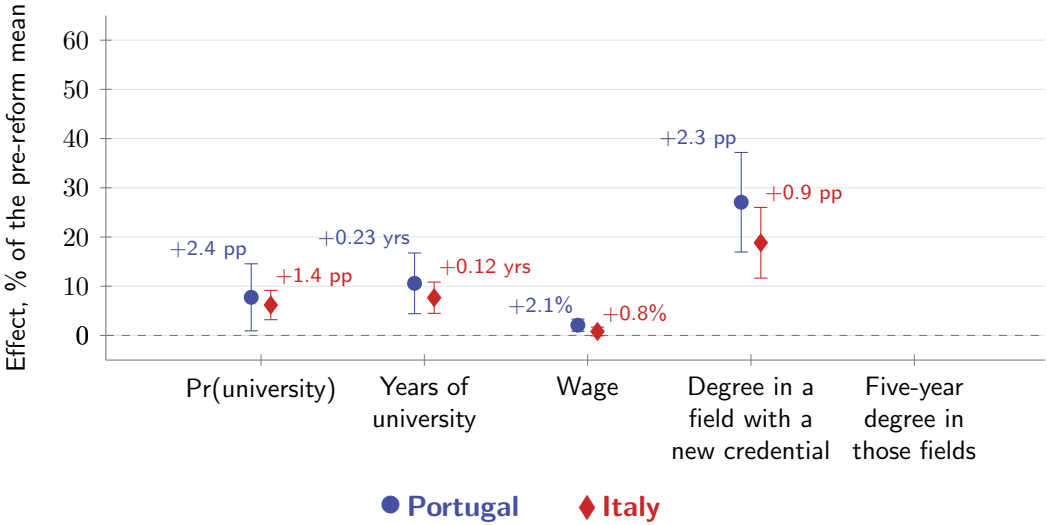
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



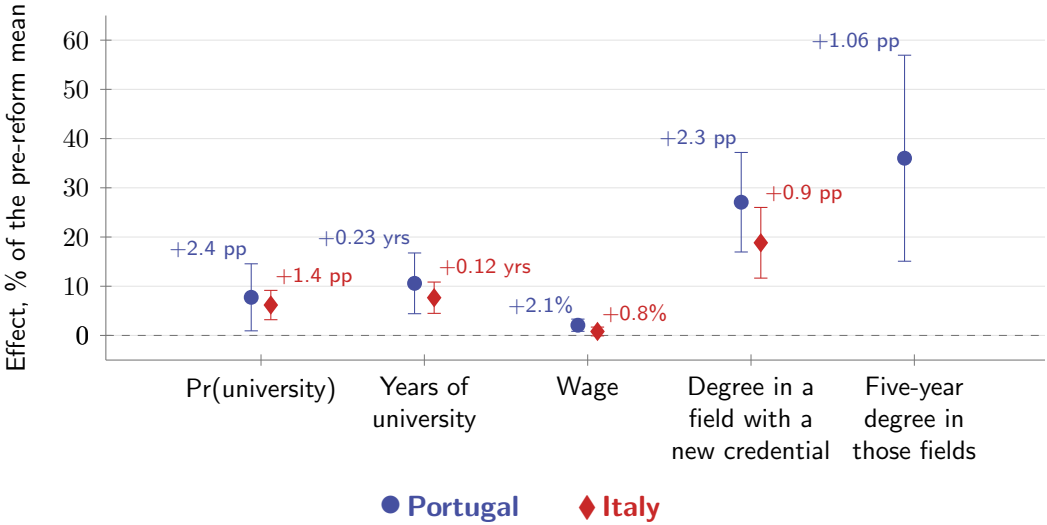
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



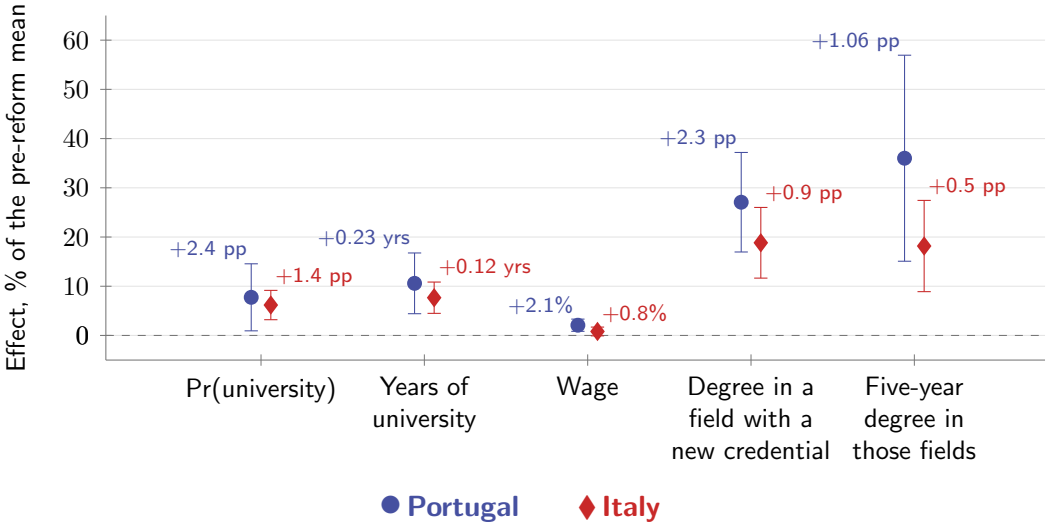
Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern



Italy: attainment and pay | Italy: fields | Italy: long degrees

Portugal and Italy: the same pattern

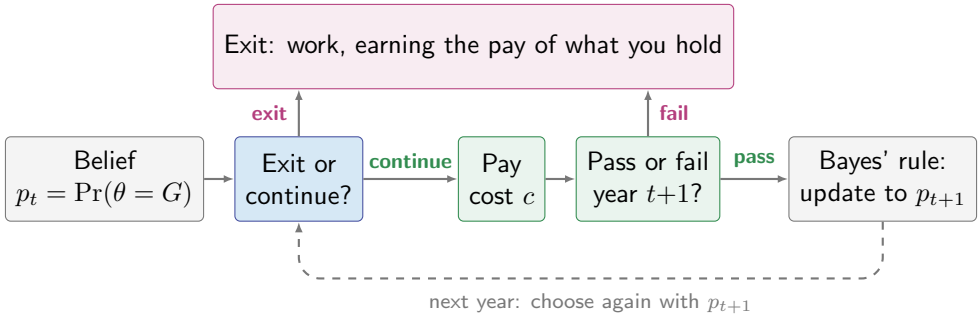


Italy: attainment and pay | Italy: fields | Italy: long degrees

Why would an exit at year 3 raise five-year completion?

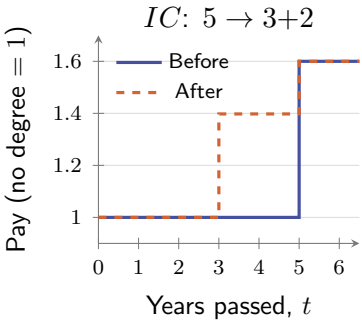
Model overview

- Match with university θ: good (G) or bad (B). Students do not know θ
- Only bad matches fail, no grades: same p_0 and t years passed, same belief



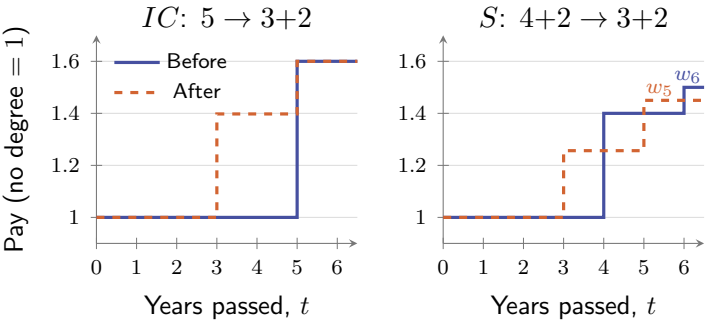
With grades

Payoffs



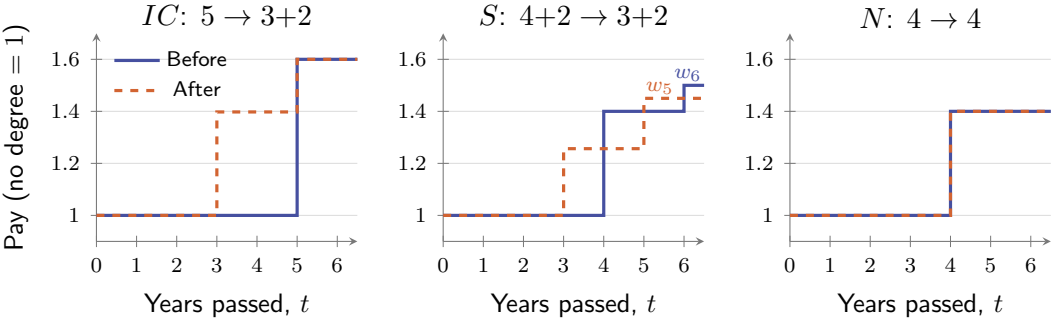
Lumpy payoff: PIAAC

Payoffs



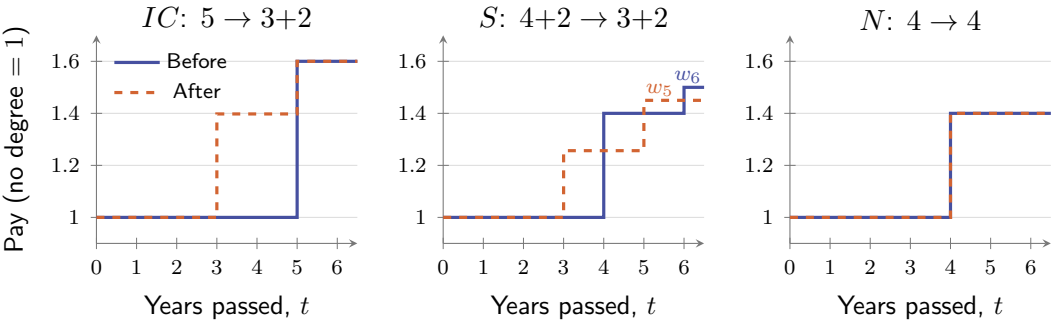
Lumpy payoff: PIAAC

Payoffs



Lumpy payoff: PIAAC

Payoffs



Lifetime pay, studying s years and then earning d a year, and its **utility**:

$$Y(s,d) = \sum_{a=s}^{T-1} \beta^a d \; - \; \sum_{a=0}^{s-1} \beta^a c, \qquad u(Y) = \frac{Y^{1-\gamma}}{1-\gamma}$$

Lumpy payoff: PIAAC

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t{+}1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1 - (1 - p_t)\, q}$$

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t{+}1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1 - (1 - p_t)\, q}$$

- **Exit or continue**, in field k, with u the utility of lifetime pay $Y(s, d_t^k)$ after s years

$$V_t^k(p_t) = \max \Big\{ \underbrace{\qquad}_{\text{exit}}, \underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{continue}} \Big\}$$

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t+1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1 - (1 - p_t)\, q}$$

- **Exit or continue**, in field k, with u the utility of lifetime pay $Y(s, d_t^k)$ after s years

$$V_t^k(p_t) = \max \Big\{ \underbrace{u\big(Y(t, d_t^k)\big)}_{\text{exit}}, \underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{continue}} \Big\}$$

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t{+}1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1 - (1 - p_t)\, q}$$

- **Exit or continue**, in field k, with u the utility of lifetime pay $Y(s, d_t^k)$ after s years

$$V_t^k(p_t) = \max \Big\{ \underbrace{u\big(Y(t, d_t^k)\big)}_{\text{exit}}, \underbrace{\overbrace{(1 - p_t)\, q}^{\Pr(\text{fail}|p_t)} + \overbrace{\big[1 - (1 - p_t)\, q\big]}^{\Pr(\text{pass}|p_t)}}_{\text{continue}} \Big\}$$

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t{+}1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1 - (1 - p_t)\, q}$$

- **Exit or continue**, in field k, with u the utility of lifetime pay $Y(s, d_t^k)$ after s years

$$V_t^k(p_t) = \max \Big\{ \underbrace{u\big(Y(t, d_t^k)\big)}_{\text{exit}}, \underbrace{\overbrace{(1 - p_t)\, q}^{\Pr(\text{fail}|p_t)}\ u\big(Y(t{+}1, d_t^k)\big) + \overbrace{\big[1 - (1 - p_t)\, q\big]}^{\Pr(\text{pass}|p_t)}\ V_{t+1}^k(p_{t+1})}_{\text{continue}} \Big\}$$

Student's choice

- **Beliefs**, with Bayes' rule: only a bad match fails, with chance q. Passing year $t{+}1$:

$$p_{t+1} = \Pr(\theta = G \mid \text{pass}, p_t) = \frac{\Pr(\text{pass} \mid \theta = G)\ \Pr(\theta = G \mid p_t)}{\Pr(\text{pass} \mid p_t)} = \frac{p_t}{1-(1-p_t)\,q}$$

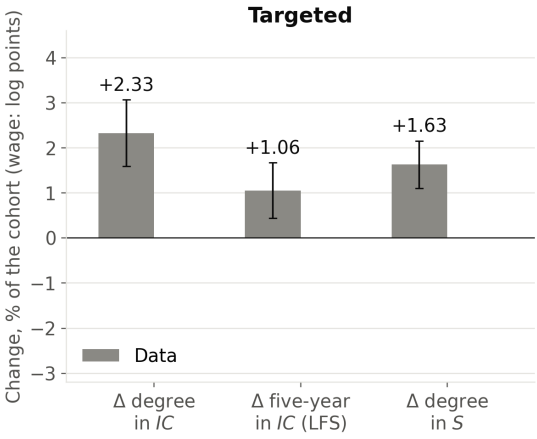
- **Exit or continue**, in field k, with u the utility of lifetime pay $Y(s, d_t^k)$ after s years

$$V_t^k(p_t) = \max\Big\{ \underbrace{u\big(Y(t, d_t^k)\big)}_{\text{exit}}, \underbrace{\overbrace{(1-p_t)\,q}^{\Pr(\text{fail}|p_t)}\, u\big(Y(t{+}1, d_t^k)\big) + \overbrace{\big[1-(1-p_t)\,q\big]}^{\Pr(\text{pass}|p_t)} V_{t+1}^k(p_{t+1})}_{\text{continue}} \Big\}$$

- **Which field, at 18**, with $v_k(p_0) = u^{-1}\big(V_0^k(p_0)\big)$ and a preference shock

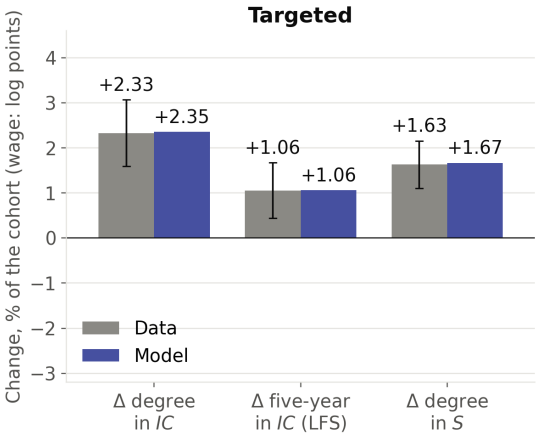
$$\max_{k \in \{W, IC, S, N\}} \Big\{ \underbrace{v_k(p_0)}_{\text{value}} + \underbrace{\mu_k}_{\text{mean pref.}} + \underbrace{\varepsilon_k}_{\text{pref. shock}} \Big\}, \qquad \varepsilon_k \sim \text{EV}(0, \sigma),\ \mu_W = 0$$

How the model fits the data



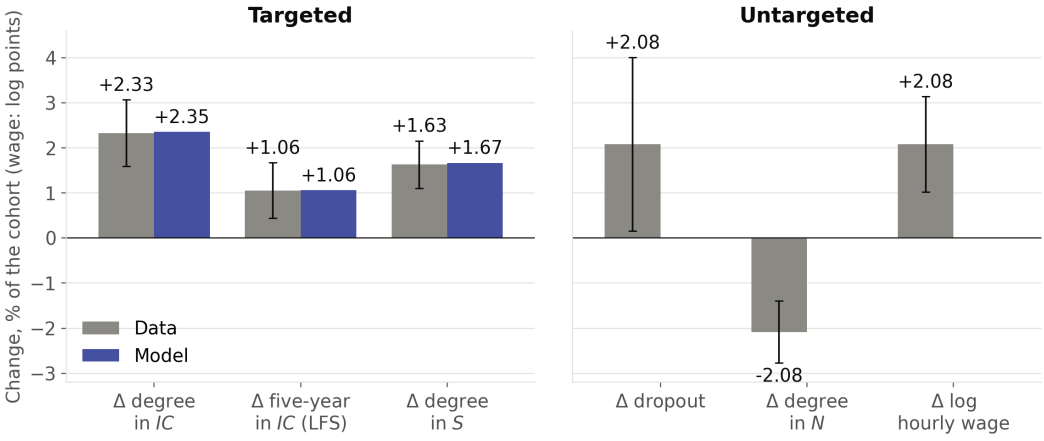
Estimation

How the model fits the data



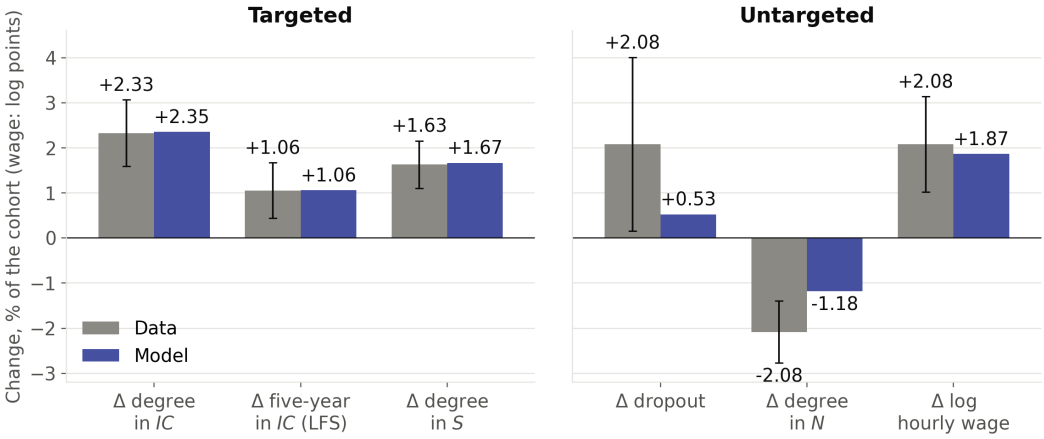
Estimation

How the model fits the data



Estimation

How the model fits the data



Estimation

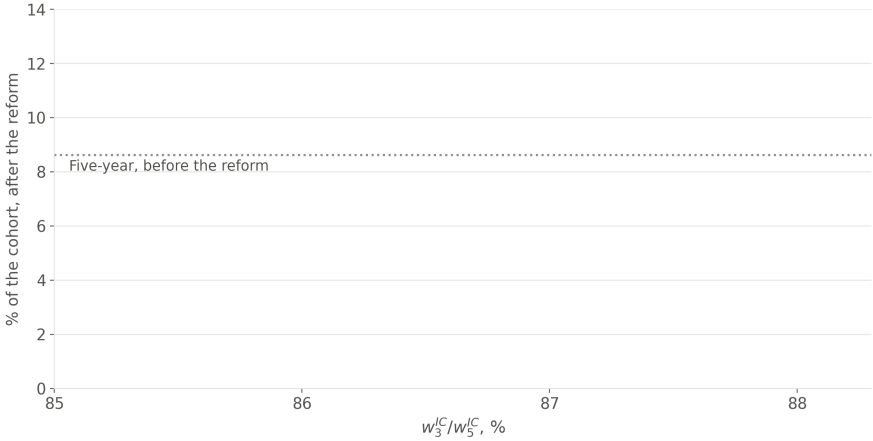
Two effects on five-year completion

- $0 \to 3{+}2$: unsure students who would not have enrolled. With a 3-year exit there is less to lose, so they try, and some go on to finish five years
- $5 \to 3$: students who would have finished five years. Still unsure after three years, they now stop with the credential

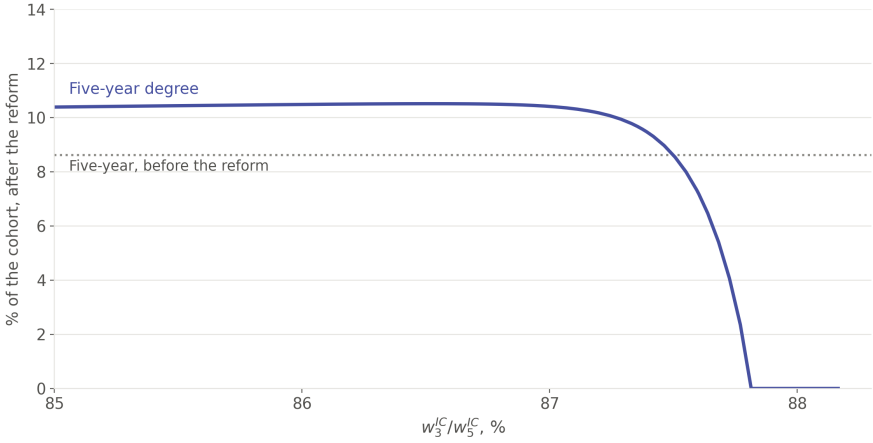
	% of the cohort
$0 \to 3{+}2$	$+1.54$
$5 \to 3$	-0.34

What can make five-year completion decrease?

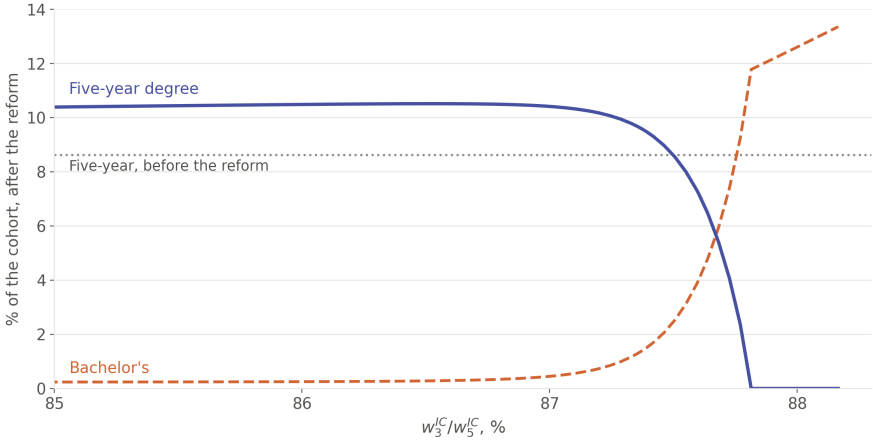
1. How much the IC pays relative to the full degree



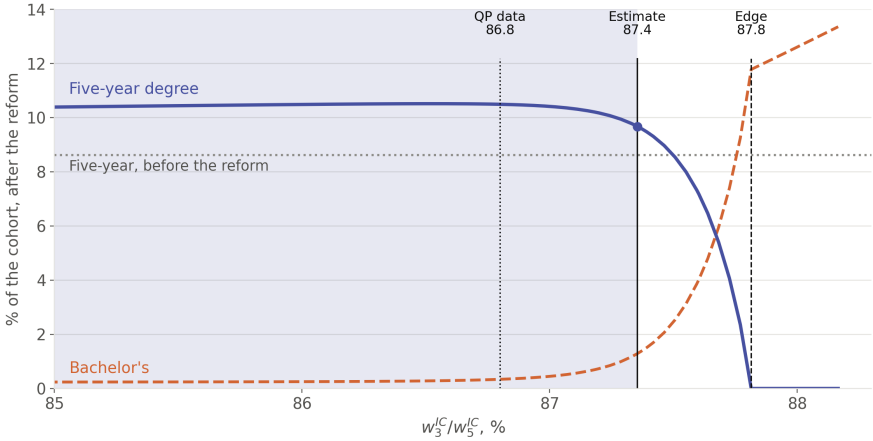
1. How much the IC pays relative to the full degree



1. How much the IC pays relative to the full degree

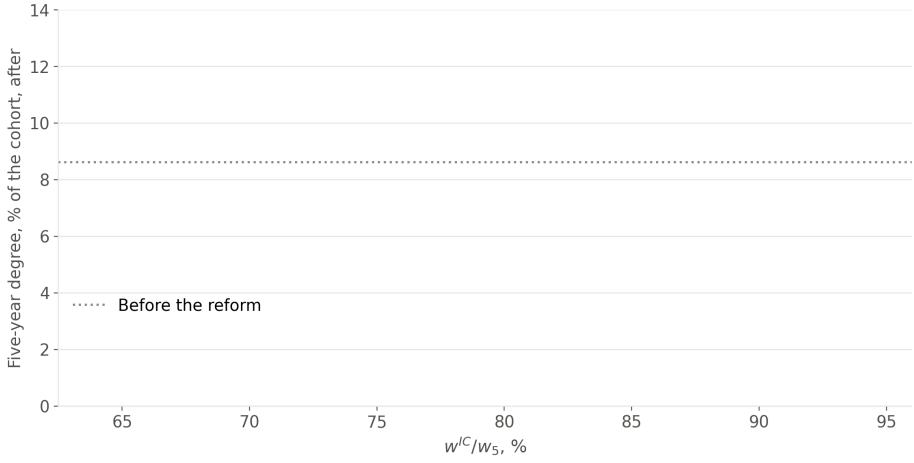


1. How much the IC pays relative to the full degree

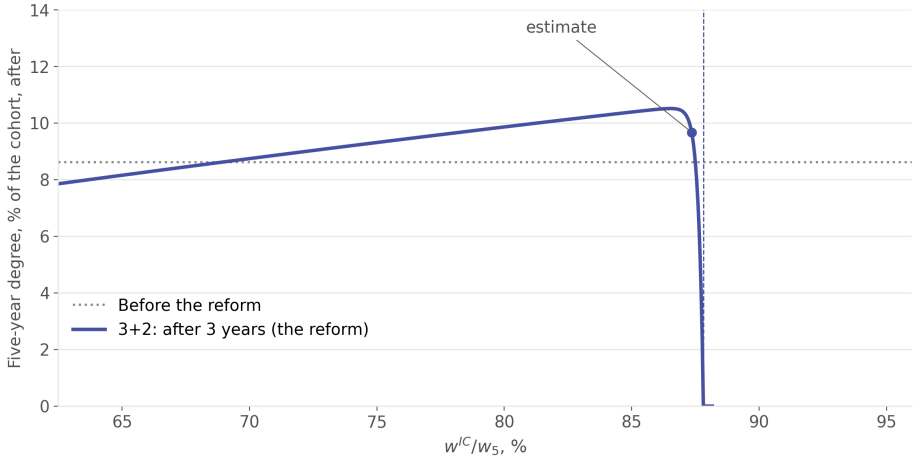


Edge: the pay at which even a sure good match stops at three years

2. How early the IC arrives

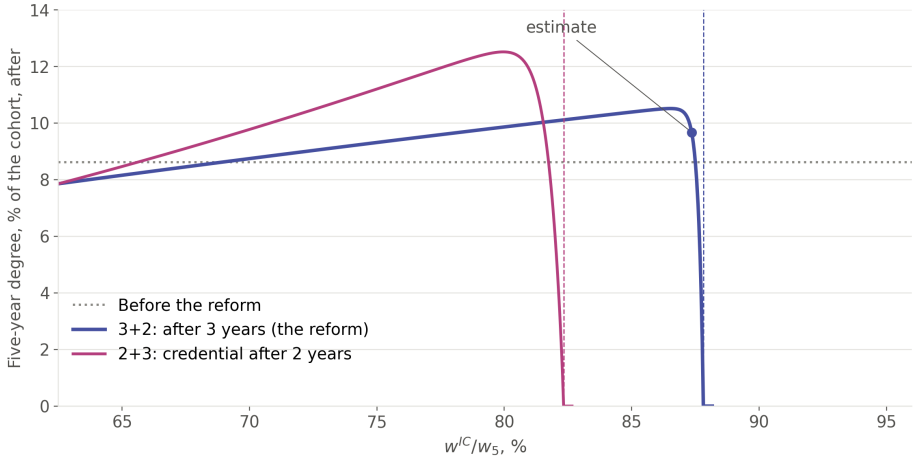


2. How early the IC arrives



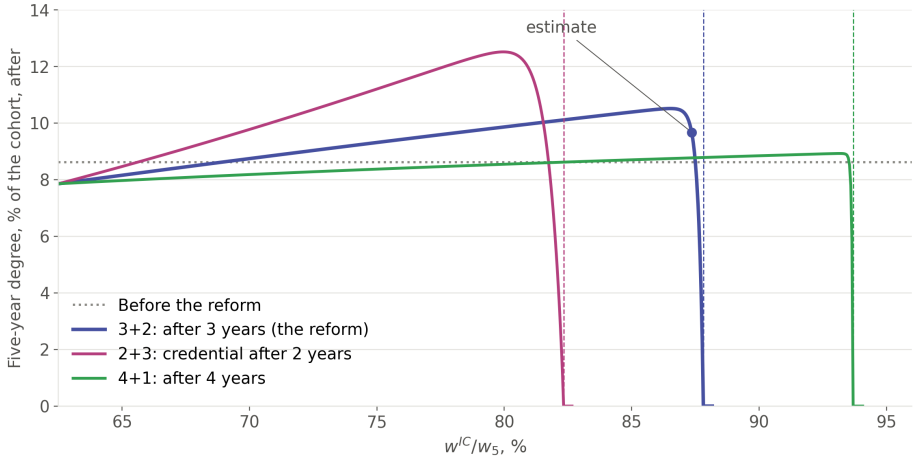
Edge (dashed): the pay at which even a sure good match stops at the credential

2. How early the IC arrives



Edge (dashed): the pay at which even a sure good match stops at the credential

2. How early the IC arrives



Edge (dashed): the pay at which even a sure good match stops at the credential

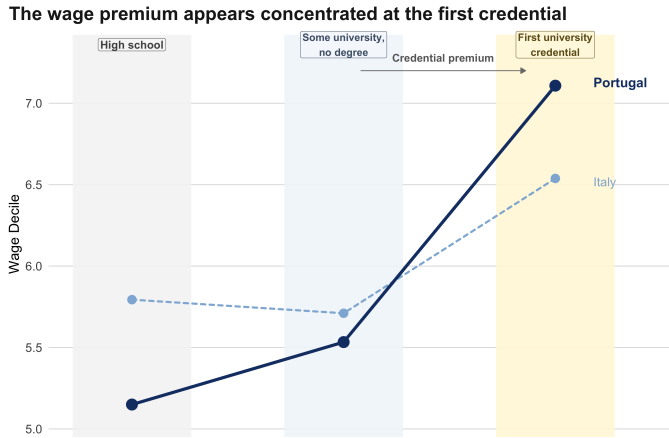
Conclusion

Bologna reform was heavily criticised

- Do shorter and more flexible degrees decrease schooling and wages?
  - No: on net, schooling and wages increase
- Does an IC discourage people from completing the full five years?
  - No: on net, more people complete the full five years when an IC is added
  - Counterfactuals show two things are important for this net effect > 0:
    1. How much it pays relative to the full degree
    2. How quickly students reach the IC ($2{+}3 \neq 3{+}2 \neq 4{+}1$)

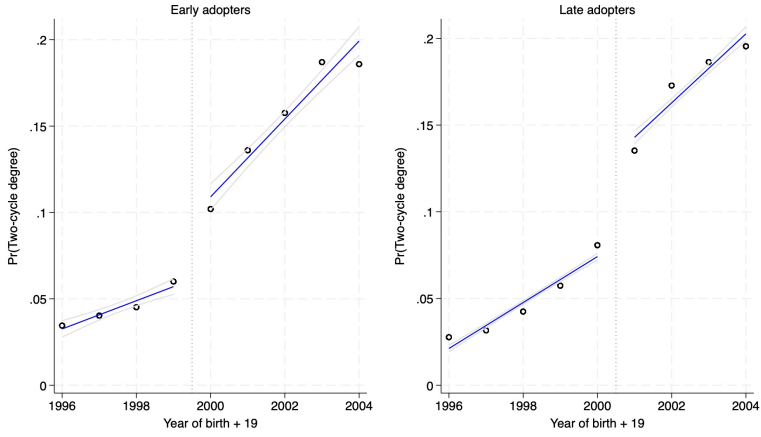
Appendix

University is risky because the payoff is lumpy



Back to the model

Italy: staggered adoption



Early adopters (2000): Piemonte, Valle d'Aosta, Friuli-Venezia Giulia, Sardinia. 84% of graduates live where they studied.

Back

What the three-year degree is worth

- **Portugal**
  - The 3-year degree keeps the name *licenciado* (Law 12-A/2008; Acórdão 3/2011)
  - 3-year engineers can register as technical engineers; the engineer title still needs 5 years

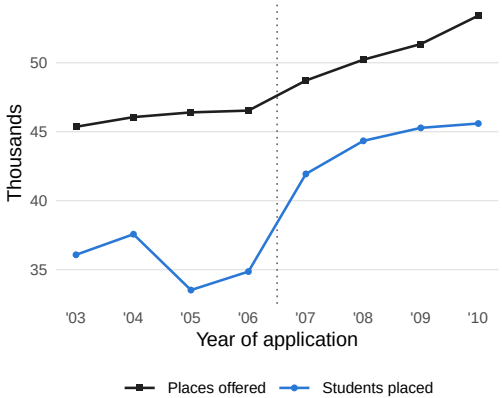
- **Italy**
  - Decree 328/2001: "junior" professional registers for 3-year graduates
  - Old *laurea* equated with the new master's

Back

Capacity is not binding; no evidence on strategic enrolment delay

Places > students placed, Portugal

Source: DGES, national access contest,



Mean age at first entry

Source: Eurostat

	Before	After
Portugal	18.84	18.88
Italy	19.19	19.18

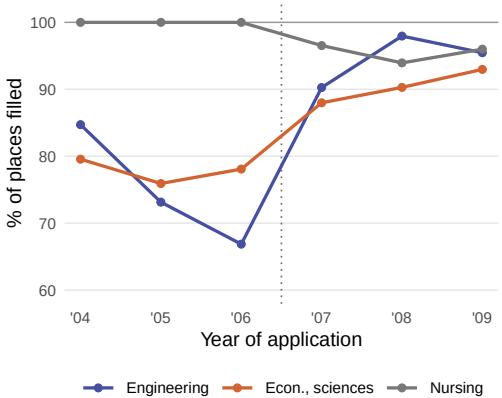
Places by field

Back

Reformed fields had empty places, nursing was full

Share of places filled, Portugal

Source: DGES, national access contest

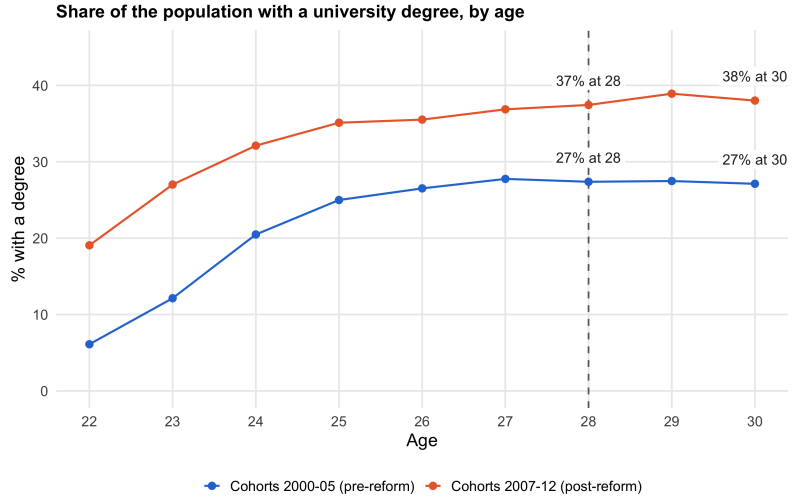


Change 2006 → 2007

	Places	Placed	% filled
Engineering	+3%	+39%	67 → 90
Econ., sciences	+1%	+14%	78 → 88
Nursing	0%	−4%	100 → 97

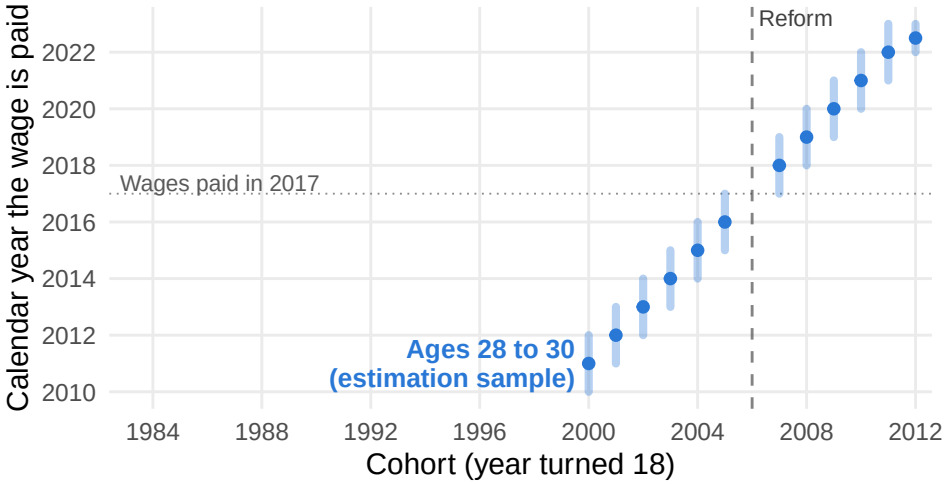
Back

Why ages 28–30: the share with a degree has nearly stopped rising



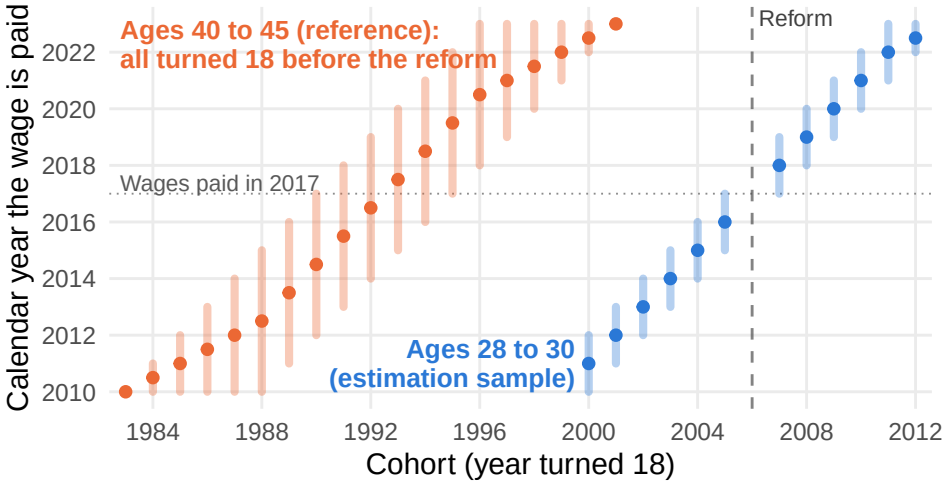
Back

Wages: relative to workers aged 40–45 paid in the same year



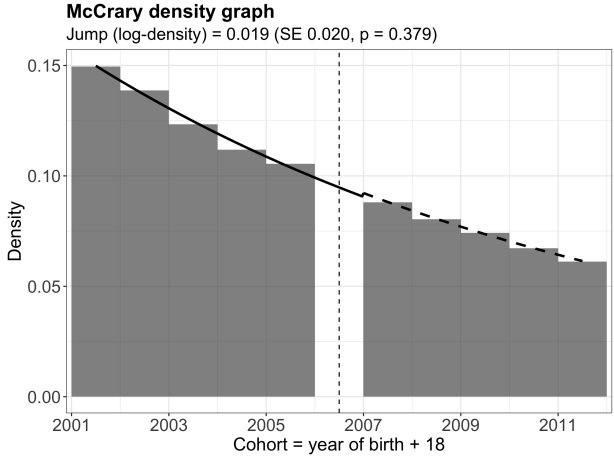
Back: RD design Back: wages

Wages: relative to workers aged 40–45 paid in the same year



Back: RD design Back: wages

Density of the running variable



Log jump 0.019 (0.020), $p = 0.379$.

Back

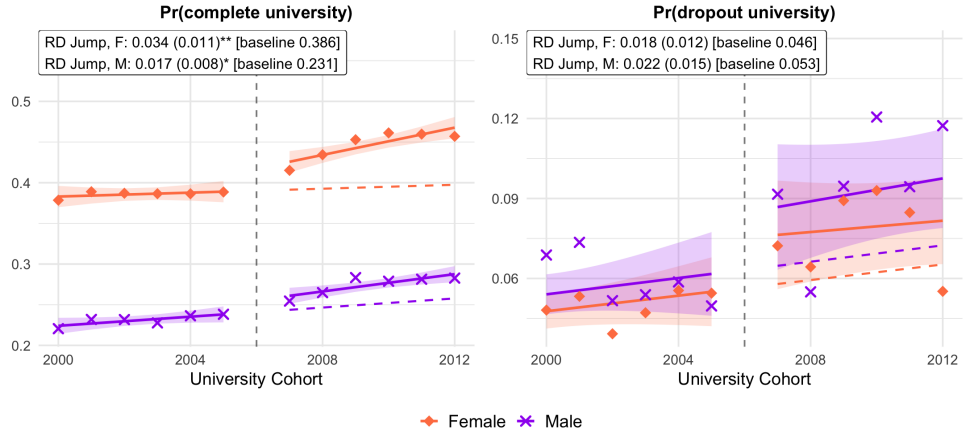
No strategic delay

Outcome: observed in employment between leaving secondary school and completing a university degree

	Whole sample	Women	Men
RD jump	0.001	0.002	−0.000
	(0.001)	(0.002)	(0.002)
Baseline	0.105	0.128	0.085

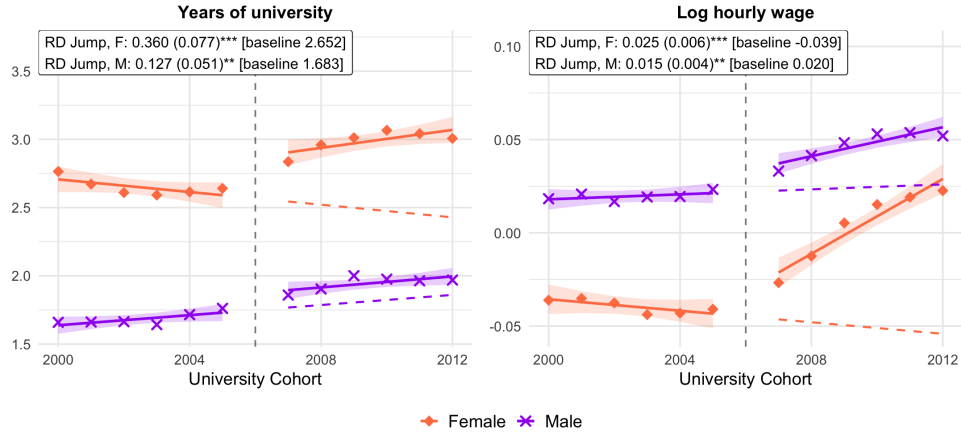
Back

Completion and dropout: women and men



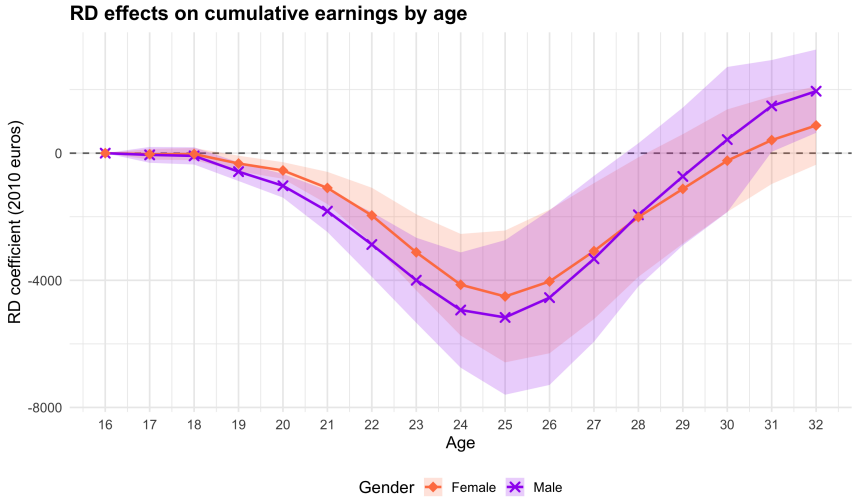
Back

Years of university and wages: women and men



Back

Cumulative earnings: women and men



Back

AKM: 46% of the wage gain comes from better paying firms

	Log hourly wage QP
All	0.021*** (0.005)
N	1,017,067

Women and men | By type of firm | Back: wages | Back: cumulative earnings

AKM: 46% of the wage gain comes from better paying firms

	Log hourly wage QP	Firm effect QP	Firm share (%)
All	0.021***	**0.010***	**46**
	(0.005)	(0.001)	
N	1,017,067	1,017,067	

- Benchmark: Portugal et al. (2024) find that 30% of the return to compulsory schooling in Portugal comes through the firm

Women and men | By type of firm | Back: wages | Back: cumulative earnings

AKM: 46% of the wage gain comes from better paying firms

	Log hourly wage QP	Firm effect QP	Firm share (%)	Pr(employed) LFS
All	0.021***	**0.010***	**46**	0.009
	(0.005)	(0.001)		(0.021)
N	1,017,067	1,017,067		42,278

- Benchmark: Portugal et al. (2024) find that 30% of the return to compulsory schooling in Portugal comes through the firm

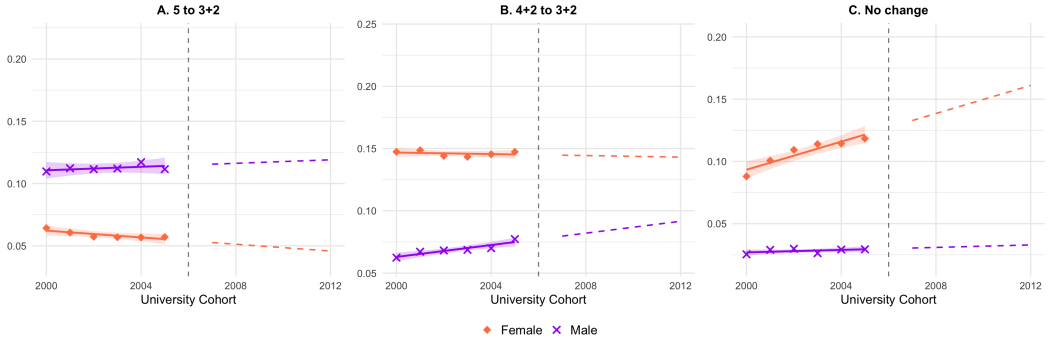
Women and men | By type of firm | Back: wages | Back: cumulative earnings

AKM: women and men

	Log hourly wage QP	Firm effect QP	Firm share (%)	Pr(employed) LFS
All	0.021***	0.010***	46	0.009
	(0.005)	(0.001)		(0.021)
Women	0.025***	0.013***	53	0.027
	(0.006)	(0.002)		(0.026)
Men	0.015**	0.005***	36	−0.011
	(0.004)	(0.001)		(0.018)
p, women = men	0.18	0.00		0.22
N	1,017,067	1,017,067		42,278

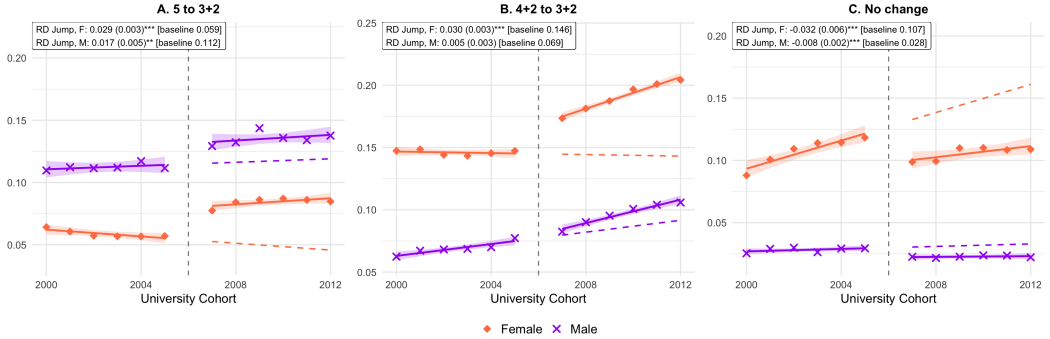
Back

Field shifts are larger for women



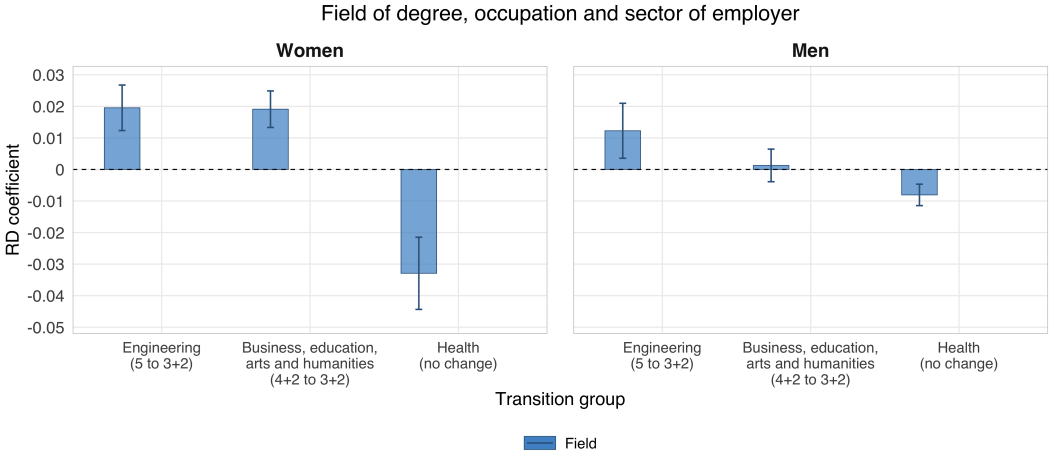
Back

Field shifts are larger for women



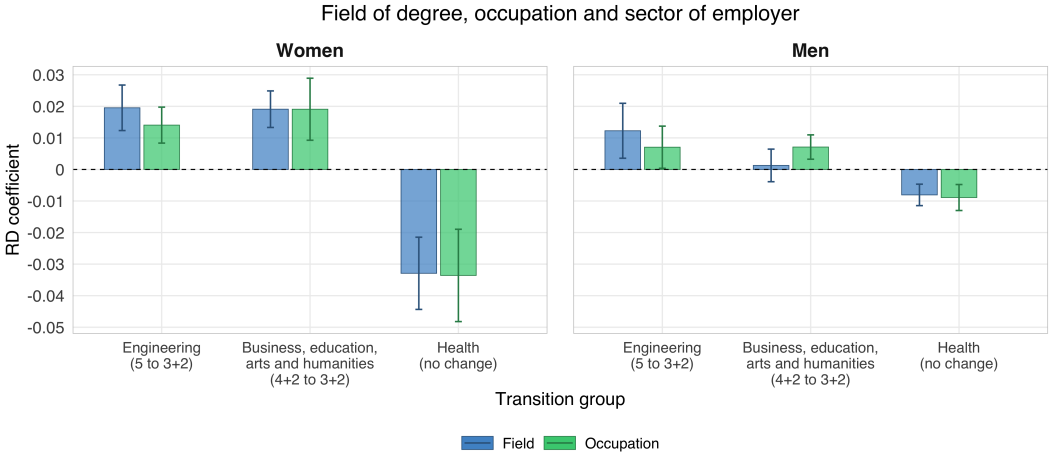
Back

By gender: larger for women, in degrees and in jobs



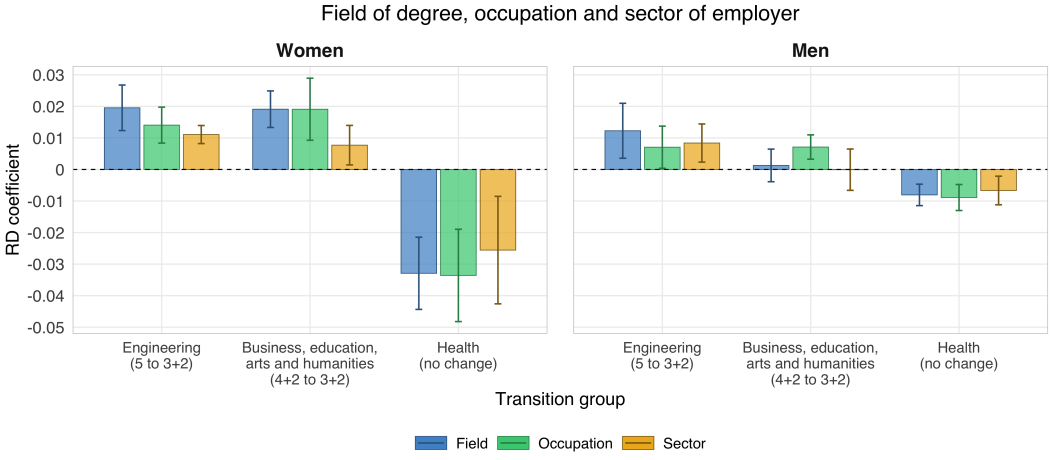
Back

By gender: larger for women, in degrees and in jobs



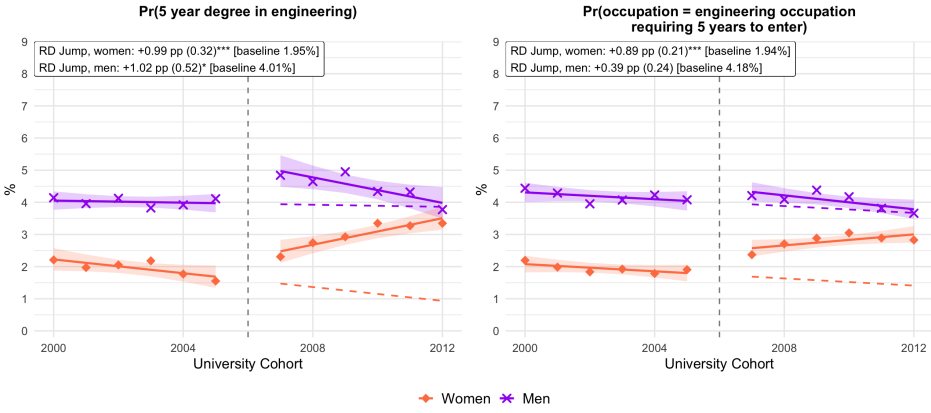
Back

By gender: larger for women, in degrees and in jobs



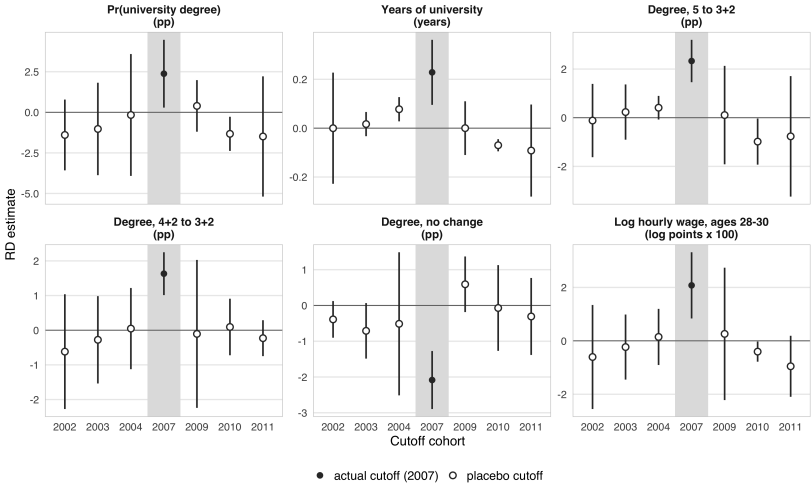
Back

By gender: more five-year engineering jobs among women



Back

Placebo cutoffs



Back: completion Back: wages Back: fields

Wages: other ages, reference groups and windows

		Ages		Reference group		Cohorts on each side	
	Main	28–29	29–30	35–45	35–40	5	4
All	0.021***	0.021***	0.024***	0.018**	0.017**	0.018***	0.013**
	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)	(0.005)	(0.003)
Women	0.025***	0.024**	0.028***	0.023***	0.022**	0.023***	0.018**
	(0.006)	(0.007)	(0.006)	(0.007)	(0.007)	(0.006)	(0.005)
Men	0.015**	0.015**	0.019***	0.012**	0.011*	0.012**	0.007**
	(0.004)	(0.005)	(0.004)	(0.005)	(0.005)	(0.004)	(0.002)
p, women = men	0.18	0.33	0.22	0.19	0.22	0.14	0.05
Workers	1,017,067	924,454	929,084	1,017,067	1,017,067	841,183	660,340

Back

Completion and years of university: QP and LFS

	University degree		Years of university	
	QP	LFS	QP	LFS
All	0.024**	0.024***	0.229***	0.126***
	(0.009)	(0.003)	(0.058)	(0.015)
Baseline	0.307	0.181	2.158	0.772
Women	0.034**	0.037***	0.360***	0.185***
	(0.011)	(0.004)	(0.077)	(0.024)
Baseline	0.386	0.229	2.652	0.938
Men	0.017*	0.012***	0.127**	0.070***
	(0.008)	(0.003)	(0.051)	(0.020)
Baseline	0.231	0.133	1.683	0.608

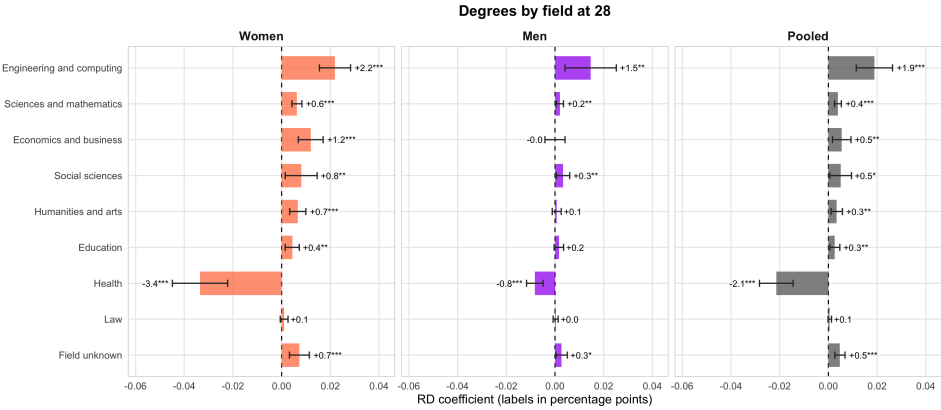
Back

Fields: QP and LFS

	5 to 3+2		4+2 to 3+2		No change	
	QP	LFS	QP	LFS	QP	LFS
All	0.023***	0.006***	0.016***	0.022***	−0.021***	−0.007***
	(0.004)	(0.002)	(0.003)	(0.002)	(0.004)	(0.001)
Baseline	0.086	0.038	0.107	0.081	0.067	0.037
Women	0.029***	0.008***	0.030***	0.037***	−0.032***	−0.010***
	(0.003)	(0.002)	(0.003)	(0.004)	(0.006)	(0.003)
Baseline	0.059	0.027	0.146	0.111	0.107	0.062
Men	0.017**	0.002	0.005	0.009***	−0.008***	−0.003**
	(0.005)	(0.002)	(0.003)	(0.002)	(0.002)	(0.001)
Baseline	0.112	0.049	0.069	0.051	0.028	0.013

Back

Degrees by detailed field



Engineering and computing: +1.9 on 7.1%, most of the rise. Health: −2.1 on 6.2%.

Back

Ministry records: applications and first-year students

		Change in share (pp)	
		Women	Men
Engineering	Applications	+2.2	+0.4
	First-year students	+2.0	+4.6
Economics and business	Applications	+2.5	+3.0
	First-year students	+3.0	+1.2
Nursing	Applications	−4.3	−1.8
	First-year students	−2.3	−1.0

Back

A tenth of workers, half of the wage gain

		Part of the effect in		
	Log wage	IT and engineering	Business and finance	All other firms
All	0.021***	0.006***	0.005***	0.010*
	(0.005)	(0.001)	(0.001)	(0.005)
Women	0.025***	0.005***	0.006***	0.014**
	(0.006)	(0.001)	(0.001)	(0.006)
Men	0.015**	0.006***	0.004***	0.004
	(0.004)	(0.001)	(0.001)	(0.004)
Share of the effect, all (%)	100	28	24	48
Share of workers, pre-reform (%)	100	4.1	5.4	90.5

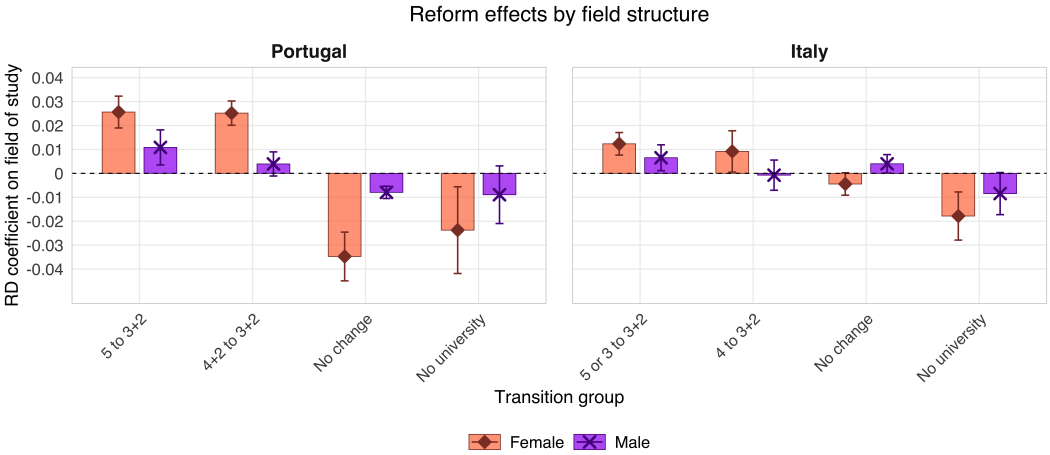
Back

Italy: attainment and pay

	Pr(university)	Years of univ.	Log net wage	Employed
All	0.014***	0.125***	0.008*	−0.002
	(0.003)	(0.026)	(0.005)	(0.004)
Baseline	0.227	1.626	7.029	0.698
Women	0.018***	0.169***	0.021***	0.002
	(0.005)	(0.039)	(0.007)	(0.005)
Baseline	0.273	1.900	6.910	0.608
Men	0.008*	0.069*	0.001	−0.003
	(0.004)	(0.035)	(0.005)	(0.005)
Baseline	0.179	1.340	7.134	0.791
p, women = men	0.17	0.06	0.02	0.55
N	269,026	269,026	142,599	269,026

Back

Italy: fields, next to Portugal



Back

Italy: engineering and healthcare degrees by length

	Engineering		Healthcare	
	Any degree	5-year degree	Any degree	5-year degree
All	0.004***	0.002*	0.006***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Baseline	0.027	0.022	0.022	0.008
Women	0.004***	0.003**	0.008***	0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
Baseline	0.012	0.009	0.030	0.009
Men	0.004	0.002	0.003*	0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
Baseline	0.043	0.034	0.015	0.006
N	269,026	269,026	269,026	269,026

Back

Key assumptions

We assume	So
Only a bad match fails, with chance q a year	Passing is always good news
Pay depends on the credential, not the match	Bad matches who pass earn the same
Preference shock ε_k is drawn once, at 18	Nobody quits out of dislike
Beliefs right on average: $\Pr(\theta = G \mid p_0) = p_0$	Students fail as often as they expect

Preference shocks Adding grades Back

Preference shocks

- At 18, each student draws a preference shock ε_k per option $k \in \{W, IC, S, N\}$
- In years of pay, like the values $v_k(p_0) = u^{-1}\big(V_0^k(p_0)\big)$ (work: pay with no degree)
- Fixed for life, unrelated to the match θ: they count at 18, not for passing or exit
- Extreme value type I, $\varepsilon_k \sim \text{EV}(0, \sigma)$, independent across options, so a logit, with mean preference μ_k ($\mu_W = 0$):

$$\Pr(k \mid p_0) = \frac{e^{(v_k(p_0)+\mu_k)/\sigma}}{\sum_{j\in\{W,IC,S,N\}} e^{(v_j(p_0)+\mu_j)/\sigma}}$$

- $\sigma \to 0$: same belief, same choice. Larger σ: preferences matter more, pay less

Back

What $v_k(p_0)$ is: the certainty equivalent

- At 18, field k is a gamble: she may pass every year, or fail and leave with what she holds
- $V_0^k(p_0)$ is the expected utility of that gamble, with belief p_0
- $v_k(p_0)$ is the sure lifetime pay that gives her the same utility:

$$u\big(v_k(p_0)\big) = V_0^k(p_0) \quad \Longrightarrow \quad v_k(p_0) = u^{-1}\big(V_0^k(p_0)\big) = \Big[(1-\gamma)\, V_0^k(p_0)\Big]^{\frac{1}{1-\gamma}}$$

- Work is not a gamble: $v_W = Y(0, w_0)$, lifetime pay with no degree
- With risk aversion ($\gamma = 2$), $v_k(p_0)$ is below the expected lifetime pay of field k. The gap is what she would pay to remove the risk
- $v_k(p_0)$ is in years of pay w_0, like the preference shocks ε_k, so the two can be added

Back

Estimation: what moves, and which data moment pins it

Parameter		Estimate	Targeted moment (% of cohort)	Data	Model
Mean preference, IC	μ_{IC}	-1.74	Degree in IC, before	8.61	8.61
Mean preference, S	μ_S	-0.62	Degree in S, before	10.67	10.67
Mean preference, N	μ_N	-0.77	Degree in N, before	6.67	6.67
Failure probability	q	0.410	Dropout, before	4.97	4.98
Scale parameter	σ	0.321	Δ degree in IC	$+2.33$	$+2.35$
Bachelor's pay	w_3^{IC}	1.3977	Δ five-year degree, LFS	$+1.06$	$+1.06$
Three-year pay	w_3^S/w_4^S	0.898	Δ degree in S	$+1.63$	$+1.67$

The distance | What is fixed | Back: student's choice | Back: fit

The estimated model: what is fixed

Parameter		Value	Where the value comes from
Discount factor	β	0.95	Stange (2012); Johnson (2013): 0.97
Risk aversion (CRRA)	γ	2	Johnson (2013); Stange (2012): 0
Working life		40 years	set by hand; Stange: to age 62
Study cost a year	c	$0.1\,w_0$	fee cap, EUR 901–949 a year
Pay, no degree	w_0	1	normalisation
Pay, IC five-year degree	w_5^{IC}	1.6	QP data
Pay, S 4-year first degree	w_4^S	1.4	QP data
Pay, S full degree	w_F^S	1.5	QP data
Pay, N degree	w^N	1.4	QP data
Starting belief	p_0	Beta(0.94, 0.91)	PISA 2003: scores, confidence

Back

The distance

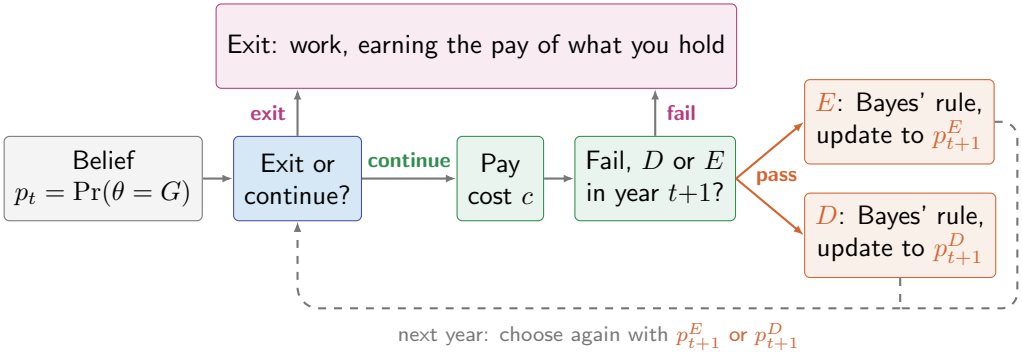
$$\hat{\psi} = \arg\min_{\psi} \sum_{m=1}^{5} \left(\frac{\hat{x}_m(\psi) - x_m}{\mathrm{SE}_m} \right)^2 \quad \text{subject to} \quad \hat{x}_k^{\text{pre}}(\psi) = x_k^{\text{pre}}, \;\; k \in \{IC, S, N\}$$

- $\psi = \left(\mu_{IC}, \mu_S, \mu_N, q, \sigma, w_3^{IC}, w_3^S / w_4^S\right)$
- Five moments x_m, % of the cohort: dropout before, and the jumps in degrees in IC, degrees in S, and five-year degrees (LFS)
- The five-year jump is a moment like the others: LFS five-year degrees, $+1.06$ (0.31)
- Distance at the estimates: 2.53

Back

Model overview with grades

Each pass now comes with a grade: excellent (E) or decent (D)



Back

Adding grades (one field)

- Years 1–4 end in a fail, or a pass with a decent (D) or an excellent (E) grade

Chance of	good match $(\theta = G)$	bad match $(\theta = B)$	given belief p_t
fail	0	q	$(1 - p_t)\, q$
pass, D	$1 - \alpha_G$	$(1 - q)(1 - \alpha_B)$	$(1 - \alpha_G)\, p_t + (1 - q)(1 - \alpha_B)(1 - p_t)$
pass, E	α_G	$(1 - q)\, \alpha_B$	$\alpha_G\, p_t + (1 - q)\, \alpha_B\, (1 - p_t)$

- Belief after the grade:

$$p^E_{t+1} = \Pr(\theta = G \mid E, p_t) = \frac{\Pr(E \mid \theta = G)\Pr(\theta = G \mid p_t)}{\Pr(E \mid p_t)} = \frac{\alpha_G\, p_t}{\Pr(E \mid p_t)}$$

$$p^D_{t+1} = \Pr(\theta = G \mid D, p_t) = \frac{\Pr(D \mid \theta = G)\Pr(\theta = G \mid p_t)}{\Pr(D \mid p_t)} = \frac{(1 - \alpha_G)\, p_t}{\Pr(D \mid p_t)}$$

- Continue: a bet on three outcomes

Back

$$V_t(p_t) = \max \Big\{ u\big(Y(t, d_t)\big),\ (1 - p_t)\, q\, u\big(Y(t{+}1, d_t)\big) + \Pr(E \mid p_t)\, V_{t+1}(p^E_{t+1}) + \Pr(D \mid p_t)\, V_{t+1}(p^D_{t+1}) \Big\}$$

References I

Angrist, J., Autor, D., and Pallais, A. (2022). Marginal effects of merit aid for low-income students. *The Quarterly Journal of Economics*, 137(2):1039–1090.

Arcidiacono, P., Aucejo, E., Maurel, A., and Ransom, T. (2025). College attrition and the dynamics of information revelation. *Journal of Political Economy*, 133(1):53–110.

Bordon, P. and Fu, C. (2015). College-major choice to college-then-major choice. *Review of Economic Studies*, 82(4):1247–1288.

Cappellari, L. and Lucifora, C. (2009). The "bologna process" and college enrollment decisions. *Labour economics*, 16(6):638–647.

Hastings, J. S., Neilson, C. A., and Zimmerman, S. D. (2013). Are some degrees worth more than others? evidence from college admission cutoffs in chile. Technical report, National Bureau of Economic Research.

Kirkeboen, L. J., Leuven, E., and Mogstad, M. (2016). Field of study, earnings, and self-selection. *The Quarterly Journal of Economics*, 131(3):1057–1111.

References II

Kroher, M., Leuze, K., Thomsen, S. L., and Trunzer, J. (2021). Did the" bologna process" achieve its goals? 20 years of empirical evidence on student enrolment, study success and labour market outcomes. Technical report, IZA Discussion Papers.

Manski, C. F. (1989). Schooling as experimentation: a reappraisal of the postsecondary dropout phenomenon. *Economics of Education review*, 8(4):305–312.

Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review*, 112(8):2580–2630.

Portugal, P., Reis, H., Guimarães, P., and Cardoso, A. R. (2024). What lies behind the returns to schooling: The role of labor market sorting and worker heterogeneity. *Review of Economics and Statistics*, pages 1–45.

Rouse, C. E. (1995). Democratization or diversion? the effect of community colleges on educational attainment. *Journal of Business & Economic Statistics*, 13(2):217–224.

Stange, K. M. (2012). An empirical investigation of the option value of college enrollment. *American Economic Journal: Applied Economics*, 4(1):49–84.

Trachter, N. (2015). Stepping stone and option value in a model of postsecondary education. *Quantitative Economics*, 6(1):223–256.